TITLE OF PROJECT-

Smart Dorm Energy Control Using CLI Reflex Agent.

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TITLE

Smart Dorm Energy Control Using Basic Command Line Reflex Agent.

INTRODUCTION

In AI, an agent is something that perceives its surroundings through sensors and takes action using actuators. It works on a basic idea: when a certain condition is given, it responds with an appropriate action. This project uses one of the simplest types of agents based on this principle.

The Smart Dorm system, designed to run in a terminal window, shows how a room can respond automatically without requiring constant human input. The idea behind it is taken from a common situation in student residential hostels often leave their rooms with lights and fans running, which wastes energy and increases electricity costs.

This system aims to solve that problem by adopting automated behaviour. It can detect when someone is present and keep things comfortable . But once the room is empty, it automatically turns off unnecessary appliances.

The goal is simple: reduce energy waste without requiring extra effort. By managing devices in the background, the system ensures energy is used only when needed.

PROBLEM STATEMENT

Honestly, we all leave switches on sometimes overnight and it wastes more energy than we realize. An empty room doesn't need cooling or heating, yet the devices are kept running, building up heat that just ends up escaping out. imagine if things adjusted automatically the moment you walked in. That alone could stop a lot of unnecessary energy wastage. Temperatures change quickly, but most device is not meant to adapt that.

Life moves on, schedules shift, people come and go—but the electricity keeps flowing ,even if there is no one in room.

Project Purpose

To design a Simple Reflex Agent using the PEAS framework.

This project runs as a dynamic simulation in Python, where factors like temperature, time, and motion are constantly changing. These inputs act as sensor readings, contin- uously feeding data into the system. As the values update in real time, they create an ongoing picture of what's happening in the room. The system stays active, processing and responding to the latest information.

At its core, the goal is to use condition–action rules to automate things like lighting and temperature control. When a certain condition is met, the system reacts instantly without any manual input.

The idea becomes easier to understand when the everyday life examples are taken into account . Over time, repeated actions reveal patterns, just like machines learn from data. Even habits can reflect the same logic behind smart systems. As repetition builds, accuracy improves just like training a model.

EXISTING METHODS

Manual Control: Relying on students to turn on or off the switches. Facts

About The Methods Mentioned:

Pros: No cost.

Unreliable to a fault - energy often cutoff without warning. Chance of inefficiency is dangerously high.

Fixed Timers: very now then, lights shuts off sharp at 9 o'clock. It does not matter if someone remains inside .No exceptions made in this flow.

Fewer options when plans shift; might feel awkward ,should study time get obstructed, like before exams.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware: Personal Computer/Laptop.

Python 3.10 or newer sits at the core. Running things happens through a terminal window. any IDE for python.

METHODOLOGY AND PURPOSE

The system has a framework that guides how the agent works. It helps the system handle things one step at a time. Each part of the system is built around goals. When you think about it in a way tasks become clear naturally. The system is designed to have boundaries that make sense.

When the room is between 18 and 28 degrees it is possible to use power. You can still be comfortable even when it is dark and the lights are on. The system can control the temperature and the lights at the time without using too much energy. The system is designed to be efficient and only uses energy when it is needed. The system is a success when the room is at a temperature and the lights are on. Saving power is just as important as having a place to live.

The system has a motion tracker that notices when someone is in the room. It also has a thermometer that checks the temperature. The system has a clock that shows the time.

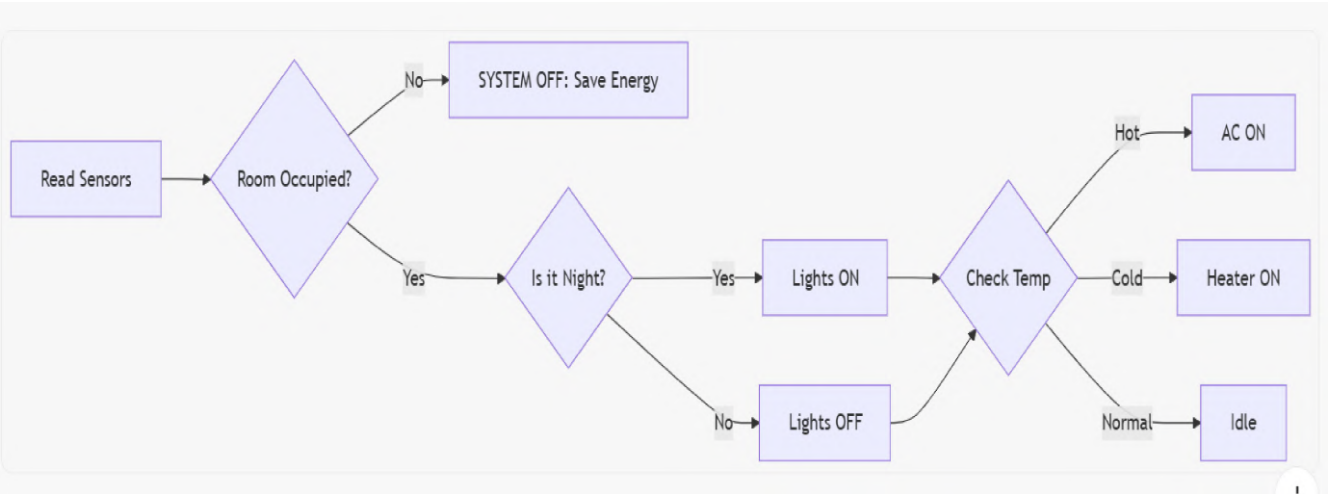
The system is designed to work on its own. When the system gets a signal from a sensor it acts away. The system notices changes. Responds to them automatically. The system is always. Responding to what is happening around it.

The system has a part that's like eyes and ears. It gets information from the surround- ings using sensors. This part of the system just collects information. Does not try to understand it. It is like an observer that is always watching what is happening around it.

The system has a brain that's like a set of rules. When something happens the system follows a set of steps. The system keeps track of what happens when certain conditions are met. The system has a set of paths that it follows to make decisions.

The system has a part that makes things happen. It is like the step, in the process. When the system decides what to do this part makes it happen.

Rough diagram of working on agent:



ALGORITHM DEVELOPMENT

The Agent Uses A Basic Reflex Pattern

Step 1: Get is occupied from the motion sensor.

Now suppose is occupied equals false - then activate the power-down rule. Every actuator turns off when that condition appears. Off happens across all units once occupancy drops out. The system responds by cutting power everywhere if no presence registers. When nothing detects movement, all actuators go inactive together. Only after vacancy confirms does every single one shut down completely.

Now suppose is occupied holds a value of True Last lights go on when clock hits seven PM till six AM. Outside those hours, they stay off. Nighttime means illuminated. Daylight shuts them down. After six in evening arrives, power flows to fixtures. Before seven at night begins, everything cuts out. Hours past midnight keep brightness active. Morning light switches it all off. When temperature passes twenty-eight degrees, air conditioning switches on.

Rule C: If Temp < 18°C, Turn Heater ON.

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When temperature passes twenty-eight degrees, air conditioning switches on. Rule C:

If Temp < 18°C, Turn Heater ON.

Step 4: Display state changes in the CLI. Step

5: Repeat.

Software Architecture Diagram

When you have a setup like this it normally connects an agent to the things around it using sensors and actuators. The agent is separate, from the environment it is working. This separation happens because of the input and output channels that connect the two. One part of the system collects information. Another makes changes. This process keeps going in a loop never stopping. . The agent and its surroundings keep working in a loop as long as the system is working.

CODING

Python

Simplified Logic Core

```
classSmartDormAgent:
    defperceive_and_act(self,env):ifnotenv.is_
        occupied:
            self.shut_down_all(env)#EnergySavingRuleelse:
            #ComfortandUtilityRulesself.manage_lights(en
            v.current_time)self.manage_climate(env.temp
            erature)
```

OUTPUT

Plaintext

Environment Change: Time: 20:00, Temp: 31°C, Occupied: True —

[AGENT ACTION]: It's dark: Lights turned ON.

[AGENT ACTION]: Too hot: AC turned ON.

Environment Change: Time: 21:00, Temp: 22°C, Occupied: False —

[AGENT ACTION]: Dorm Empty: All systems powered down.

Key System Implementations:

When nobody is in the room, that rule steps ahead of everything else its job is stopping power waste. What matters most? Cutting energy use where there is wastage of it , without any purpose.

A screen full of live updates shows each sensor and what it's doing. One line at a time tracks how parts respond. Text flashes as values shift without delay. Every device gets its own row updating constantly. You see numbers change the moment they do. Lines scroll when new readings arrive. Each piece talks through letters on black background. Nothing waits everything appears now.

SIGNIFICANT PROJECT RESULTS

When nobody is in the room the rule to save energy is the important thing. Its main job is to stop wasting power. What is most important to us is to reduce energy use where it is being wasted for no reason.

A screen shows us what is happening with each sensor now. It shows us one thing at a time like how it is moving. The words on the screen change soon as the values change. Every device has its line on the screen that is always updating. The lines, on the screen move down when we get information. Each part of the system communicates with us by showing letters on screen. Nothing is delayed. We see everything now. The energy saving rule is what matters most in this situation. It is always working to stop power waste.

Testing and Refinement:

Testing is going on with some conditions.

When it got really hot and nobody was there the air conditioner did not turn on. The test was successful.

It was midnight. Even though it was late the room was still occupied. The lights and the cooling system were working. Everything was working as it should. The system passed the test without any problems.The project can be used in real life.

Even though we are testing it on a computer it can be used in places like dorm rooms with things, like Raspberry Pi or Arduino to control devices. How does it work? It works by sending signals to turn things on or off instead of just pretending to. Testing with stress parameters and the air conditioner and the cooling system and the lights are all part of this project. The project and the testing and the air conditioner and the cooling system are all working well.

Limits of the System:

A student comes to the room around six. Each time the student comes it is like the time for the Simple Reflex Agent. The Simple Reflex Agent does not remember when

the student came before. It only knows the student is there when the student enters the room. the student visits the Simple Reflex Agent starts fresh.

A classroom can be very quiet. The Simple Reflex Agent might turn off. This happens because the Simple Reflex Agent gets a little up. It thinks the room is empty when it is just quiet. Sometimes nobody is moving,. The Simple Reflex Agent thinks nobody is there. The Simple Reflex Agent makes mistakes, like this.

CONCLUSIONS

A working dorm room setup puts fundamentals of AI and ML ideas into action. When basic AI follows the right structure, thanks to PEAS- it handles real tasks well. Energy control improves without complex tools. Smarter choices happen through clear design, not extra power.

FUTURE ENHANCEMENTS

Model-Based Agent: Adding an internal state to track user habits.

Out here, command line gives way to a browser interface. Access opens up through visual tools instead of typed entries. A screen appears where steps once stayed hidden in scripts. Watching progress feels simpler now, guided by layout rather than memory. Clicks replace commands. What ran in darkness steps into view, frame by frame.

REFERENCES

Russell, S. J., & Norvig. Artificial Intelligence: A Modern Approach.

CSA2001 Course Modules, VIT Bhopal University.